COMAP AI Use Disclosure Report

HiMCM 2019 Problem A — Terminal Charging Kiosks

Team GoHiMCM

This report documents artificial-intelligence assistance used while preparing our contest materials, in accordance with COMAP contest rules on AI disclosure. No AI system invented numerical results. All tables and figures come from local Python DES runs that write `results/*.csv` and `paper_figures/*.png`.

1 Tools and scope

Tool	Role	Contested artifacts touched
Ollama (`qwen2.5:7b`), local	Drafted pack proposals inside the LangGraph reflection loop (`propose`→`validate`→`simulate`→`reflect`→`check`)	`src/des_agent/reflection_agent.py`; `results/reflection_*.json`; `results/autonomous_planning_summary.json`
LangGraph	Orchestrated the self-reflection control flow and planning subtasks	`src/des_agent/planning_agent.py`
Cursor (AI coding agent)	Assisted SimPy queue state-machine coding and FastMCP tool wrappers	`src/des_env/airport_sim.py`; `src/mcp_server/server.py`; `src/bdi_agents/passenger_bdi.py`
Cursor	Assisted Pareto densify scripts and figure wiring	`src/optimization/pareto_sizing.py`; `paper_figures/`
Cursor	Assisted LaTeX document shells and path wiring	`paper/*.tex`

Interaction prompts that affected model or algorithm design were logged in `prompts/cursor_log.md` for auditability.

2 Local generation of simulation data

1. All DES KPIs were produced on a local Python 3.10+ environment (SimPy + NumPy/Pandas/Matplotlib).
2. FastMCP tools call the same in-process DES functions used by scripts; they do not fetch external contest data.
3. Ollama runs locally as `qwen2.5:7b`. If Ollama is offline, a deterministic rule proposer continues. The DES engine still executes every candidate pack.
4. Locked evidence includes `stage2_baseline_metrics.csv`, `optimal_kiosk_sizing.csv`, `thunderstorm_stress` and the 300 DPI figures under `paper_figures/`.

3 What the human team did independently

We attest that the following were conceived, specified, and signed off by the student team:

1. **Scientific design:** BDI passenger agent states, SimPy resource pools, CapEx/area caps, storm injection rules, and Pareto objectives.
2. **Acceptance criteria:** mean wait ≤ 5 minutes and reneging probability ≤ 0.05.
3. **Summary Sheet:** narrative priorities and final one-page claims. Every numeric entry was typed from measured CSV outputs after human inspection. The Summary Sheet was completed independently by human teammates.
4. **Airport authority letter:** operational SOP language (three-minute walk grid; free first 15 minutes on the fast kiosk; storm lane checklist) was completed independently by human teammates. AI assistance, if any, was limited to LaTeX layout only.

4 What AI assistance was used for

- Drafting SimPy process boilerplate and FastMCP tool signatures.
- Driving LangGraph reflection / planning loops via local Ollama.
- Generating initial LaTeX table/figure wiring tied to existing CSV and PNG filenames.
- Fixing pytest paths and small typing/docstring gaps.

5 What AI was not used for

- Inventing or editing locked KPI cells in `results/`.
- Fabricating storm or Pareto statistics for the Summary Sheet.
- Submitting without human review of every reported number.

6 Attestation

We confirm that this disclosure is complete to the best of our knowledge and that contest claims rest on locally executed simulation outputs.

Team GoHiMCM Date: _____